Steve Gwade

Fullstack AI Engineer

Yaounde, Cameroon · Open to Relocation

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Professional Summary

Fullstack AI Engineer with 3+ years of experience building and deploying web applications integrated with machine learning models. Develops asynchronous server architectures (FastAPI, Django), type-safe frontends (React, TypeScript, Vite), and model orchestration layers (LLM pipelines, multi-modal applications, and vector search indices). Experienced engineering background event streams, writing automated test suites, and managing containerized runtime environments for low-latency production applications.

TECHNICAL SKILLS

- **Languages & Protocols:** Python (Asyncio), TypeScript, JavaScript, SQL (PostgreSQL, SQLite), WebSockets, REST, Bash.
- **Frontend Engineering:** React, Vite assembly toolchains, Next.js, Redux, Tailwind CSS, Frontend State Management.
- **Backend & Infrastructure:** FastAPI, Django, Flask, Nginx, Redis Caching layers, Asynchronous Task Queues, Microservices.
- **AI Application Engineering:** LLMs, Multi-Modal Systems, LangChain, Structured Data Parsing (Pydantic validation).
- **Machine Learning & Execution:** PyTorch, Vector Search Systems (FAISS, HNSW indexing), TensorRT INT8 Edge Calibration, CUDA.
- **DevOps & Observability:** Docker, Docker Compose, GitHub Actions CI/CD pipelines, AWS (EventBridge, SQS, EC2, S3), MLflow.
- **Methodologies:** Agile/Scrum Sprint Lifecycles, Unit & Integration Testing, API Contract Verification, Competitive Programming.

WORK EXPERIENCE

Fullstack AI Engineer — Apr 2026 – Present
MITS SARL – Moabi B2B Travel Agency — *Yaounde, Cameroon*

- Built an asynchronous LLM orchestration layer using FastAPI and structured Pydantic validation to handle real-time WhatsApp conversational streams, reducing 99th-percentile API response latency to under 120ms.
- Implemented responsive, multi-turn chat views in React and TypeScript with localized context management, backed by a distributed Redis and DynamoDB caching layer handling 500+ concurrent user streams.
- Decoupled heavy model inference components from core application web servers using event-driven AWS EventBridge and SQS queues, reducing cloud resource provisioning costs by 35%.
- Mentored 2 junior engineers while setting up unified GitHub Actions CI/CD workflows to execute parallel PyTest and Vitest suites, maintaining over 92% test coverage across 15+ core system modules.

Fullstack AI Solutions Engineer — Jan 2025 – Jan 2026
JIANTS – Systems Intelligence Lab — *Douala, Cameroon*

- Deployed 4 multi-modal pipelines (Vision, OCR, STT, TTS) inside React/TypeScript evaluation dashboards to expose real-time model metrics to enterprise clients.
- Refactored PyTorch tensor operations and data-loading layers using MLflow tracking, reducing pipeline feature ingestion bottlenecks by 25%.
- Quantized local computer vision layers via TensorRT INT8 post-training protocols, boosting model inference speed from 18 to 28 FPS on embedded CUDA platforms.
- Audited runtime memory footprints to implement tensor activation checkpointing, reclaiming 40% VRAM during concurrent batch execution cycles.

Fullstack Software Engineer *(Part-time)* — Mar 2024 – Dec 2024
TGNix – IT Consulting — *Douala, Cameroon*

- Integrated predictive scoring modules into 2 ERP web applications using React frontends, cutting manual administrative review times by 30%.
- Optimized PostgreSQL queries and built composite database indexes across 10M+ transaction records, accelerating execution times by 25%.
- Engineered non-blocking FastAPI backend routes for distributed calculation jobs, handling 150k+ daily API updates without message dropping or connection timeouts.
- Containerized multi-tiered application environments using Docker and Docker Compose, reducing local developer onboarding spin-up times from 4 hours to 15 minutes.

AI Backend Engineer *(Internship)* — Jun 2023 – Nov 2023
AITECAF – AI Technologies for Africa — *Yaounde, Cameroon*

- Wrote RESTful Python APIs serving 5 internal classification models, maintaining 99.95% system uptime across a 6-month tracking window.

- Designed automated text parsing and embedding generation layers using LangChain, scaling document extraction infrastructure to process 50k+ tokens per second.
- Fine-tuned a custom YOLOv8 object detection model architecture, optimizing anchor box allocations to achieve a sub-5ms processing speed and a 4.2% increase in mAP@0.5.
- Automated data engineering ingestion pipelines for 10k+ high-resolution images, using stratified spatial hashing to eliminate evaluation data leakage.

SELECTED PROJECTS

Kite – Cross-Device Scratchpad App *Open Source, 2025*

- Built a cross-device workspace web app pairing a React frontend (Vite and TypeScript) with an asynchronous FastAPI WebSocket server.
- Designed an offline-first data reconciliation mechanism using client-side state diffing, maintaining document parity for 1,200+ active concurrent sessions.
- Implemented batched REST synchronization endpoints, reducing total network payloads by 20% during peak usage periods.

VERO – Fullstack Workspace for Grounded AI *Open-Source Project, 2025*

- Engineered a document-search RAG platform using a React UI, indexing 100k+ multi-format document blocks with context retrieval responses under 45ms.
- Configured a hybrid search pipeline combining dense vector embeddings with BM25 keyword tracking, optimized using Reciprocal Rank Fusion (RRF).

LINA – Assistive Multi-Modal AI System for the Blind *Huawei ICT Competition 2025*

- Co-developed a multi-modal assistive application integrating web camera streams to deliver frame updates at 30+ FPS over WebSocket interfaces.
- Trained binary face anti-spoofing classifiers to secure application access control lines, successfully blocking 99% of visual spoofing vectors.
- **Awarded 1st Place Worldwide (Innovative Track)** out of thousands of global entrants for code-base performance, execution speeds, and model compression.

PlumVision – Computer Vision Quality Control System *JCIA Hackathon Winner 2025*

- Built a 3MB ConvNeXt image classification service using representation learning, reaching a 98% accuracy rating across production test samples.
- Wrote a monitoring web interface using React and FastAPI to stream analytics, completing frame execution cycles safely within a 12ms target window.

Multi-Task Transfer Learning for Malaria Diagnosis *MSc Research Thesis, 2025*

- Formulated a joint multi-task deep learning problem in PyTorch executing simultaneous semantic cell segmentation and parasite detection from blood smears.
- Evaluated negative transfer optimization dynamics across distinct model heads, establishing a stable joint topology yielding a 97% macro F_1-score.
- Published a reproducible open-source research repository featuring pipeline configuration matrices, execution scripts, and verified pre-trained model weights for public access.

EDUCATION

MSc in Engineering (Data Science & AI) 2023 – 2025
National Higher Polytechnic School of Douala *Douala, Cameroon*

- Graduation Status: Graduated with Distinction (Top 5% of cohort, **GPA**: 3.6/4).
- Core Focus: Distributed Systems Architecture, Applied Deep Learning Engineering, High-Scale Data Processing Pipelines.
- **Platform Source Code**: github.com/GwadeSteve/MTTL-Research-Malaria

BSc in Engineering (Computer Science) 2020 – 2023
National Higher Polytechnic School of Douala *Douala, Cameroon*

- Core Modules: Object-Oriented Design, Problem Solving, Algorithmic Complexity analysis, Advanced Data Structures, Relational Database Engineering.

AWARDS & RECOGNITION

- **1st Place Worldwide Winner**, Huawei ICT Competition 2025 (Innovative Track) – Core Developer for the LINA Assistive Vision project.
- **Ranked 2nd Across Africa**, Orange Digital Center Champions League 2025 – Algorithmic and Competitive Programming tournament (CodinGame).
- **Winner, Best AI Solution** for Optimization and Efficiency, JCIA 2025 Track – Awarded for PlumVision real-time throughput and deployment speed.
- **Winner, Most Innovative AI Prototype**, JCIA 2025 Track – Awarded for EatWise, a multi-modal framework for targeted nutritional parsing.